S&S Quiz-11 solution

Sol: 1(a) – Given a stable LTI system function

A* 1/2

$$H(z) = \frac{1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2}}{z^{-1}\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)} \quad -(1)$$

Poles at
$z^{-1} = 0 \Rightarrow z = \infty$
$1 - \frac{1}{2}z^{-1} = 0 \Rightarrow z = \frac{1}{2}$
& $1 - \frac{1}{3}z^{-1} = 0 \Rightarrow z = \frac{1}{3}$

By Partial fraction, we need to simply H(z)

So, $H(z) = \frac{A}{z^{-1}} + \frac{B}{1-\frac{1}{2}z^{-1}} + \frac{C}{1-\frac{1}{3}z^{-1}}$

$$\Rightarrow \frac{1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2}}{z^{-1}\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)} = \frac{A}{z^{-1}} + \frac{B}{1-\frac{1}{2}z^{-1}} + \frac{C}{1-\frac{1}{3}z^{-1}}$$

$$\Rightarrow 1 - \frac{4}{3}z^{-1} + \frac{1}{2}z^{-2} = A\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right) + B\left(z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right) + C\left(z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right) \quad -(2)$$

Now Put $z^{-1} = 0$, we get

$$\Rightarrow 1 - \frac{4}{3}\times(0) + \frac{1}{2}(0) = A\left(1-\frac{1}{2}\times 0\right)\left(1-\frac{1}{3}(0)\right) + B\times 0\left(1-\frac{1}{3}z^{-1}\right) + C(0)\left(1-\frac{1}{2}z^{-1}\right)$$

$$\Rightarrow 1 = A\times 1\times 1$$

So, $\boxed{A = 1}$

Now Put $z^{-1} = 2$, we get

$$\Rightarrow 1 - \frac{4}{3}\times 2 + \frac{1}{2}\times 2\times 2 = A\left(1-\frac{1}{2}\times 2\right)\left(1-\frac{1}{3}\times 2\right) + B\times 2\left(1-\frac{1}{3}\times 2\right) + C\times 2\left(1-\frac{1}{2}\times 2\right)$$

$$\Rightarrow 1 - \frac{8}{3} + 2 = A(1-1)\left(1-\frac{2}{3}\right) + 2B\left(1-\frac{2}{3}\right) + C\times 2(1-1)$$

$$\Rightarrow 3 - \frac{8}{3} = A\times 0 + 2B\times\left(\frac{3-2}{3}\right) + C\times 2\times 0$$

P.T.O

$$\Rightarrow \frac{9-8}{3} = 2B \times 1/3$$

$$\Rightarrow 1/3 = 2B \times 1/3$$

So, $\boxed{B = 1/2}$

Now put $\bar{z}^1 = 3$, we get

$$\Rightarrow 1 - \frac{4}{3} \times 3 + \frac{1}{2} \times 3 \times 3 = A\left(1 - \frac{1}{2} \times 3\right)\left(1 - \frac{1}{3} \times 3\right) + B \times 3\left(1 - \frac{1}{3} \times 3\right) + C \times 3\left(1 - \frac{1}{2} \times 3\right)$$

$$\Rightarrow 1 - 4 + \frac{9}{2} = A(1 - 3/2) \times 0 + 3 \times B \times 0 + C \times 3 \times \left(1 - \frac{3}{2}\right)$$

$$\Rightarrow -3 + \frac{9}{2} = 3C \times \left(\frac{2-3}{2}\right)$$

$$\Rightarrow \frac{-6+9}{2} = \frac{3}{2} = 3C \times (-1/2)$$

(1)

$$\Rightarrow \frac{3}{2} = -3 \times C$$

So, $\boxed{C = -1}$

Now put the value of A, B & C in equation (1), we get

$$\boxed{H(z) = \frac{1}{\bar{z}^1} + \frac{1/2}{1 - \frac{1}{2}\bar{z}^1} - \frac{1}{1 - \frac{1}{3}\bar{z}^1}}$$

From A**:

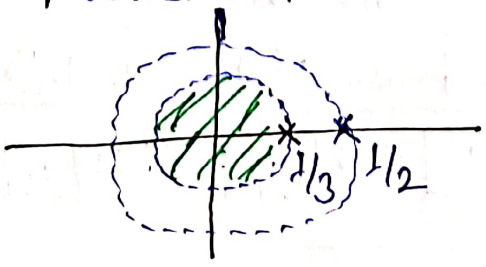
Poles

$z = \infty$, $z = 1/2$ & $z = 1/3$

Region of Convergence

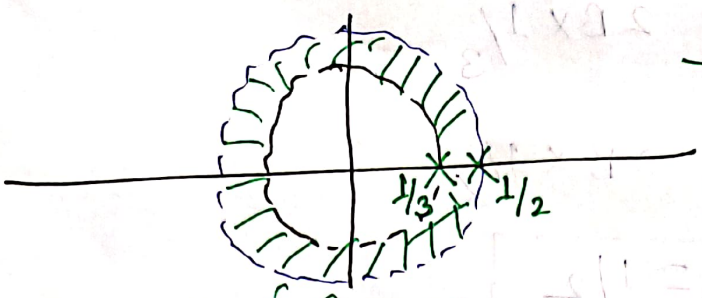
→ There are three possible ROC

(a)



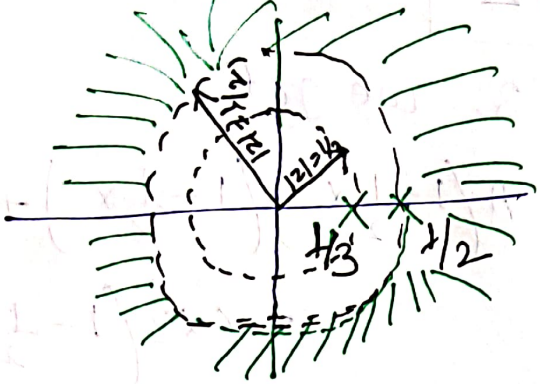
$\boxed{0 < |z| < 1/3}$

(b)



$\frac{1}{3} < |z| < \frac{1}{2}$

(c)



$|z| > \frac{1}{2}$ but excludes $|z| = \infty$

Since it is given that the LTI system is stable, the ROC should include unit circle ie $|z|=1$. Hence it implies that the overall ROC is $|z| > \frac{1}{2}$ but excludes $|z| = \infty$.

→ (1) mark

Impulse Response $h[n]$

→ $$X(z) \triangleq \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} \qquad x[n] \xleftrightarrow{z} X(z)$$

We know that

→ $\alpha^n u[n] \xleftrightarrow{z} \frac{1}{1-\alpha z^{-1}}$ $|z| > |\alpha|$ (1 mark)

→ $x[n-n_0] \xleftrightarrow{z} z^{-n_0} X(z)$ with ROC = R, except for the possible addition & deletion of the origin or infinity

→ $\delta[n] \xleftrightarrow{z} 1$ (1/2 mark)

→ $\delta[n-1] \xleftrightarrow{z} \sum_{n=-\infty}^{\infty} \delta[n-1] \cdot z^{-n} = z^{-1}$

→ $$\delta[n+1] \xleftrightarrow{z} \sum_{n=-\infty}^{\infty} \delta[n+1] \cdot z^{-n} = z$$

Now the impulse response of $H(z)$ is $h[n]$

So, $$h[n] = \delta[n+1] + \frac{1}{2} \cdot \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[n]$$ (1 mark)

Causality

→ A discrete-time LTI system is causal if & only if the ROC of its system function is the exterior of a circle, including infinity.

→ Here the ROC is the exterior of a circle (i.e $|z| > 1/2$) but it excludes the infinity. so this is not Causal.

or,

→ Here one Pole is sitting at $z = \infty$, so this system is not Causal.

↳ (1) mark



Sol: 1(b) Given stable LTI system

$$H(z) = \frac{z - 1/2}{z^2 + \frac{z}{2} - \frac{3}{16}} = \frac{P(z)}{Q(z)} \quad \text{--- (1)}$$

Here $Q(z) = z^2 + \frac{z}{2} - \frac{3}{16}$, this is Quadratic Polynomial in the form of $P(x) = ax^2 + bx + c = (x+\alpha)(x+\beta)$

poles when $H(z) = \frac{N(z)}{D(z)}$, $D(z) \neq 0$

Here $a = 1$, $b = 1/2$ & $c = -\frac{3}{16}$

we know that from quadratic polynomial the solution of x is α & β. i.e root are α & β.

& hence;

$$\alpha = \frac{-b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} \quad \& \quad \beta = \frac{-b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}$$

Here $D = \sqrt{b^2 - 4ac}$

$$= \sqrt{\left(\frac{1}{2}\right)^2 - 4 \times 1 \times \left(\frac{-3}{16}\right)}$$

$$= \sqrt{\frac{1}{4} + \cancel{4} \times \frac{3}{\cancel{16}4}}$$

$$= \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$\boxed{D = \sqrt{\frac{\cancel{4}1}{4}} = \sqrt{1} = 1}$$

So, $\alpha = \frac{-b}{2a} + \frac{D}{2a}$

$$= \frac{-1/2}{2 \times 1} + \frac{1}{2 \times 1}$$

$$= -\frac{1}{4} + \frac{1}{2}$$

$$\boxed{\alpha = \frac{-1+2}{4} = \frac{1}{4}}$$

$$\beta = \frac{-b}{2a} - \frac{D}{2a}$$

$$= \frac{-1/2}{2\times 1} - \frac{1}{2\times 1}$$

$$= -\frac{1}{4} - \frac{1}{2}$$

$$= \frac{-1-2}{4}$$

$$\boxed{\beta = -\frac{3}{4}}$$

So, $\Theta(z) = z^2 + \frac{z}{2} - \frac{3}{16}$ ~~$= (z+\frac{1}{4})(z$~~

$$\boxed{\Theta(z) = \left(z-\frac{1}{4}\right)\left(z+\frac{3}{4}\right)}$$

So, $H(z) = \dfrac{z-\frac{1}{2}}{\left(z-\frac{1}{4}\right)\left(z+\frac{3}{4}\right)}$ — (2) ⟹ Gives poles at $z = 1/4$ & $z = -3/4$ (1/2)

By Partial fraction

$$H(z) = \frac{z-1/2}{\left(z-\frac{1}{4}\right)\left(z+\frac{3}{4}\right)} = \frac{A}{z-\frac{1}{4}} + \frac{B}{z+\frac{3}{4}}$$

$$\Rightarrow z - 1/2 = A\left(z+\frac{3}{4}\right) + B\left(z-\frac{1}{4}\right)$$

Put $z = 1/4$

$$\Rightarrow \frac{1}{4} - \frac{1}{2} = A\left(\frac{1}{4}+\frac{3}{4}\right) + B(0)$$

$$\Rightarrow \frac{1-2}{4} = A \times \left(\frac{4}{4}\right)$$

$$\Rightarrow \frac{-1}{4} = A$$

So,

$$\boxed{A = -\frac{1}{4}}$$

Put $z = -\frac{3}{4}$

$$\Rightarrow -\frac{3}{4} - \frac{1}{2} = A\left(-\frac{3}{4} + \frac{3}{4}\right) + B\left(-\frac{3}{4} - \frac{1}{4}\right)$$

$$\Rightarrow \frac{-3-2}{4} = A \times 0 + B\left(\frac{-3-1}{4}\right)$$

$$\Rightarrow \frac{-5}{4} = B \times \frac{-4}{4}^{1}$$

$$\Rightarrow -B = -5/4$$

So, $\boxed{B = 5/4}$ (1)

Now, Put the value of A & B in equation (2), we get

$$H(z) = \frac{-1/4}{z - 1/4} + \frac{5/4}{z + \frac{3}{4}} \quad \text{(3)}$$

$$H(z) = H_1(z) + H_2(z) \quad \text{(4)}$$

Here, $H_1(z) = \dfrac{-1/4}{z - 1/4}$

Divide Numerator & Denominator by z, we get

$$H_1(z) = \frac{(-1/4) \times \frac{1}{z}}{\frac{1}{z}\left(z - \frac{1}{4}\right)}$$

$$H_1(z) = \frac{-\frac{1}{4} z^{-1}}{1 - \frac{1}{4} z^{-1}} \quad \text{(5)}$$

$$H_2(z) = \frac{5/4}{z + 3/4}$$

→ Divide Numerator & Denominator by z, we get

$$H_2(z) = \frac{5/4\, z^{-1}}{1 + 3/4\, z^{-1}} \quad -(6)$$

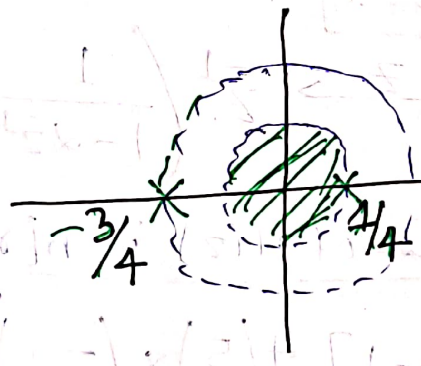
Putting $H_1(z)$ & $H_2(z)$ Value in equation (4), we get

$$H(z) = \frac{-1/4\, z^{-1}}{1 - 1/4\, z^{-1}} + \frac{5/4\, z^{-1}}{1 + 3/4\, z^{-1}} \quad -(7)$$

Region of Convergence

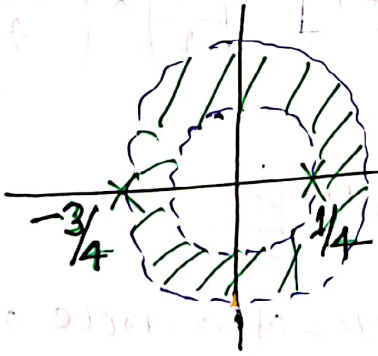
→ There are three possible ROC

(a)



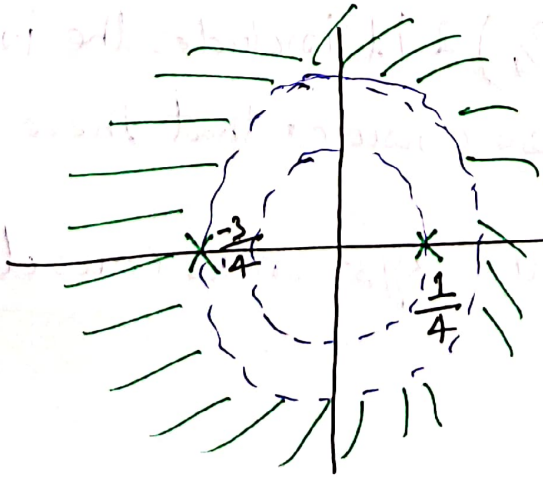
$0 < |z| < 1/4$

(b)



$\frac{1}{4} < |z| < \frac{3}{4}$

(c)



$|z| > \frac{3}{4}$

→ since it is given that the LTI system is stable, the ROC should include unit circle i.e $|z|=1$. Hence it implies that the overall ROC is $\boxed{|z| > \frac{3}{4}}$ → option C (9)

→ 1 mark

Impulse Response h[n]

We know that

→ $x[n-n_0] \xleftrightarrow{Z} z^{-n_0} X(z)$ with ROC = R, except for the possible addition or deletion of the origin & infinity

→ $\alpha^n u[n] \xleftrightarrow{Z} \frac{1}{1-\alpha z^{-1}}$ $|z| > |\alpha|$

→ $\alpha^{n-n_0} \cdot u[n-n_0] \xleftrightarrow{Z} z^{-n_0}\left(\frac{1}{1-\alpha z^{-1}}\right)$ (½)

Put $n_0 = 1$

→ $\alpha^{n-1} \cdot u[n-1] \xleftrightarrow{Z} z^{-1}\left(\frac{1}{1-\alpha z^{-1}}\right)$

Now the impulse response of H(z) is h[n] (1)

$$\boxed{h[n] = \left(-\frac{1}{4}\right)\times\left(\frac{1}{4}\right)^{n-1} \cdot u[n-1] + \left(\frac{5}{4}\right)\times\left(-\frac{3}{4}\right)^{n-1} \cdot u[n-1]}$$

Causality

→ Here the ROC is the exterior of a circle outside the outermost pole ($|z| > 3/4$) & it includes the infinity. This impulse response also ensures that there is no pole sitting at $z = \infty$.

So this system is Causal. → 1 mark